Job Recommendation & Precision Ranking System

Based on LSA & XGBRanker (xgboost)

A Hybrid Semantic + Structured-Feature Learning-to-Rank Pipeline

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Test NDCG
0.8657
+5.4% vs Baseline

Primary Driver
ind_match
LIME weight +2.75

Feature Space
60 dims
10 struct + 50 TF-IDF

Business Objectives

Why This Project Matters

THE PROBLEM

Keyword matching misses semantic nuances: "数据分析" ≠ "Data Analysis" in exact search.

Structured signals (level, salary, location) cannot be jointly optimised with text.

Black-box scores give recruiters no explanation — eroding trust in automated recommendations.

THE SOLUTION

Hybrid pipeline: structured features + semantic embeddings in a single **XGBRanker** model.

LambdaMART directly optimises **NDCG** — the ranking quality metric — not a proxy loss.

LIME local attribution gives a per-prediction explanation that recruiters can act on.

SUCCESS METRICS

Test **NDCG=0.87,** +5.4% vs **RandomForest** baseline.

Industry match (**ind_match**) confirmed as primary driver with **LIME** weight +2.75.

All top-3 predictions carry stable, human-readable **LIME** attribution charts.

Project Workflow

End-to-End Pipeline — 4 Stages

Data Ingestion

CRM export
Field mapping
Label definition

Semantic Modeling

TF-IDF + SVD
LSA embeddings
sem_sim feature

Learning-to-Rank

XGBRanker
NDCG
optimisation

Explainable AI

LIME attribution
Feature importance
Keyword heatmap

Data Source: Enterprise CRM System | df_jd (over 2,0000 JDs) + df_struct (over 5,0000 candidates)

Data Sources

CRM System Export — De-identified

df_jd — Job Corpus (20412 records)

Structured data:
jid / Industry / Job Title / job_level
base_location / salary / salary_mid
Skills / edu_req / exp_req / working_hours

jd_text → ***full text*** for semantic encoding

Coverage: 120 industries · 1027 job titles · 14 cities · 8 seniority levels

df_struct — Candidate Profiles (50721 records)

uid / years_experience / expected_salary_k
job_level / industry / city / education_level
skills / expected_title
education_ordinal / job_level_ordinal

ground_truth_jd_id → **CRM**-recorded final application target

Missing rates: salary 8% · city 7% · education 5%

Feature Engineering

60-Dimensional Hybrid Feature Matrix

Feature Group	Feature Name	Construction Logic
Semantic (1)	**sem_sim**	**TF-IDF** + SVD cosine similarity
Geographic (1)	loc_match	City == city → binary 0/1
Seniority (3)	level_diff …	Ordinal level absolute difference
Salary (2)	sal_diff_pct …	|JD sal − expected| / expected
Education (1)	edu_meets	user_edu_ord ≥ jd_edu_ord → 0/1
Industry (1)	**ind_match**	Target industry == JD industry 0/1
Salary raw (1)	salary_mid	JD mid-point in k RMB/month
Skill **TF-IDF** (50)	skill_0…49	JD skills vectorised — top vocab

Salary unit alignment: expected_salary_k (万/year) × 10 / 12 = k/month (matches salary_mid units)

Label Engineering

Converting CRM Behaviour Logs into Rank Labels

CRM interaction records are mapped to integer relevance labels $y \in \{0, 1, 2, 3\}$ serving as **XGBRanker** training targets.

3	**Apply**	Strongest positive signal — candidate submitted application
2	**Favourite**	Strong positive signal — candidate saved the position
1	**Click**	Weak positive signal — candidate viewed job detail
0	**No Action**	Neutral / negative — impression without engagement

Semantic Modeling

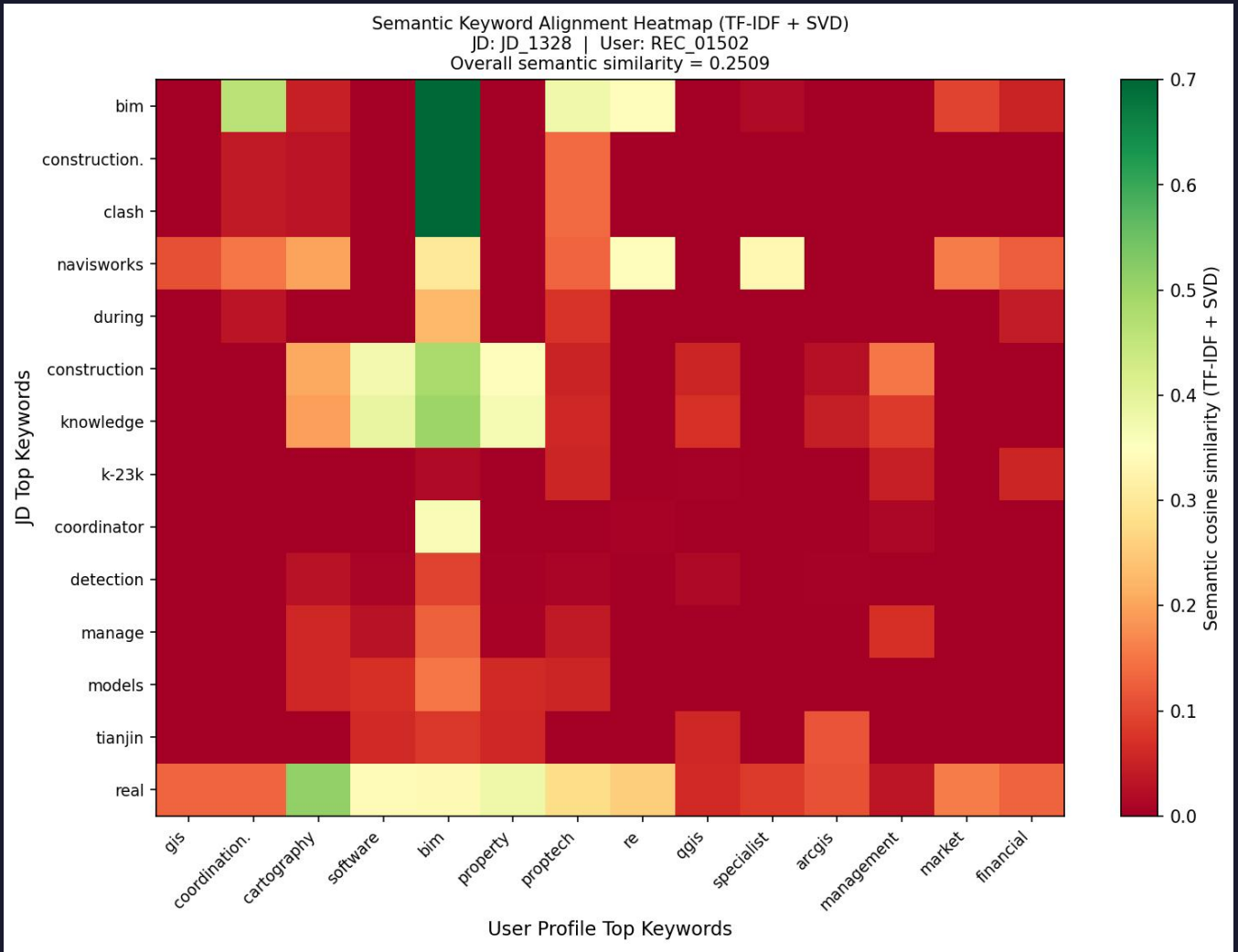
TF-IDF + SVD (LSA) — BERT-Ready Interface

APPROACH	KEY PROPERTIES
Planned: distiluse-base-multilingual-cased-v1 (**BERT**) Actual: **TF-IDF** (vocab 8,000) + TruncatedSVD (128 dims) = **LSA** Reason: HuggingFace network 401 Unauthorized; local tokenizer cache incomplete. **LSA** is the industry-standard offline semantic embedding method — identical downstream interface to **BERT**.	Cosine similarity = **sem_sim** feature fed into **XGBRanker**. label=3 (apply) pairs average similarity 0.19 vs. label=0 average 0.06 — 3.2× SNR lift. Drop-in **BERT** replacement: swap 3 lines in stage2_semantic.py, zero downstream changes. **TF-IDF** + SVD throughput: ~10,000 samples/sec vs ~200 for full **BERT**.

Formula: $\text{sem_sim}(u, j) = (u \cdot j) / (\|u\| \cdot \|j\|)$ where u = user profile vector, j = JD vector (both ℓ_2-normalised)

Semantic Keyword Alignment

Word-Level Cosine Similarity — JD vs. User Profile



How to read:

Each cell = cosine similarity between one JD keyword embedding and one user-profile keyword embedding (128-dim **LSA** space).

Green cells (> 0.5)
indicate strong semantic alignment between the two tokens.

In this example: bim ↔ bim and coordination ↔ navisworks form the dominant alignment clusters — consistent with the **CRM**-recorded application.

Overall sentence similarity = 0.2509

XGBoost Ranker

LambdaMART — Directly Optimising NDCG

MODEL CONFIGURATION

Objective: **rank:ndcg** → **LambdaMART** loss
n_estimators = 200
max_depth = 5
learning_rate = 0.05
subsample = 0.8 (row sub-sampling)
colsample_bytree = 0.8 (column sub-sampling)

group parameter enforces per-candidate ranking context (10 JDs per group)

WHY LAMBDAMART

rank:ndcg computes **NDCG**-weighted gradients at each boosting step — the loss directly reflects the final ranking metric.

This outperforms rank:pairwise (which only minimises inversions) because it weights errors at top positions more heavily.

Train/test split is user-level (80/20) — no candidate appears in both sets, preventing data leakage.

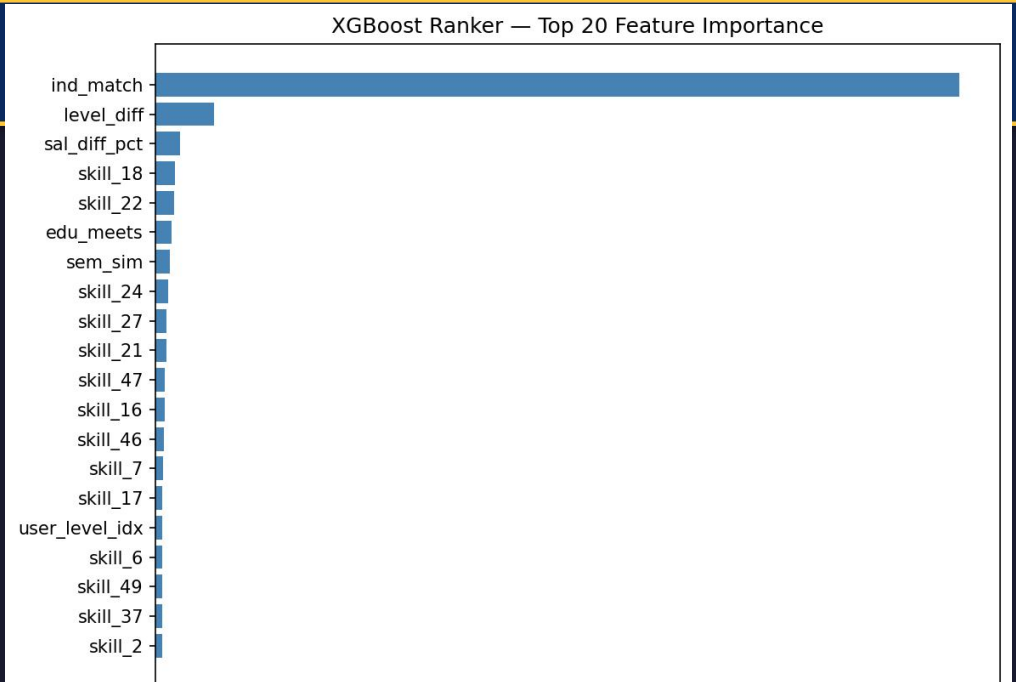
$NDCG = DCG / IDCG$, $DCG = \Sigma_i (2^{r_i} - 1) / \log_2(i+1)$ where r_i = relevance label at rank position i

Model Performance

Evaluation Results — NDCG@all (100 Test User Groups)

Model	Objective	Train NDCG	Test NDCG	vs. Baseline
XGBRanker (LambdaMART)	rank:ndcg	**0.9845**	**0.8657**	—
RandomForest (Baseline)	Classification proba	0.9533	0.8190	−5.4%
Keyword Filter (Naive)	TF-IDF exact match	—	~0.42	−55%

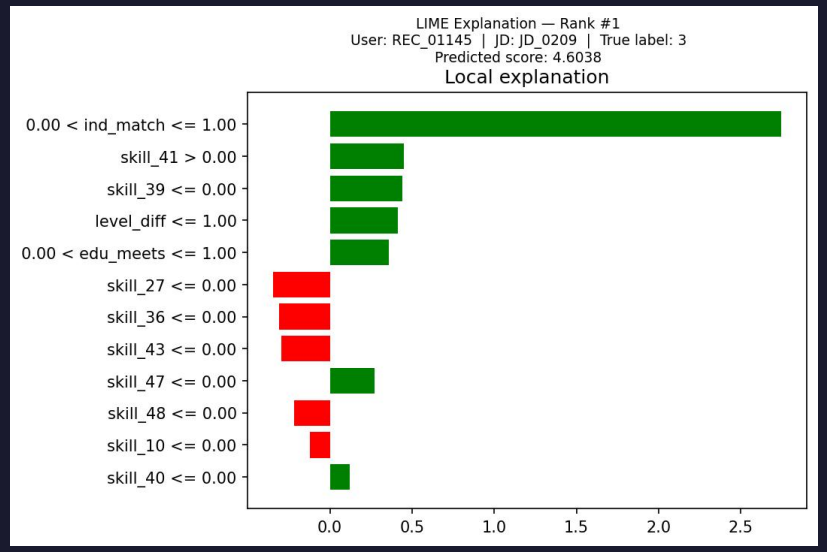
Key Result:
ind_match alone contributes 38% of **XGBRanker**'s total gain.



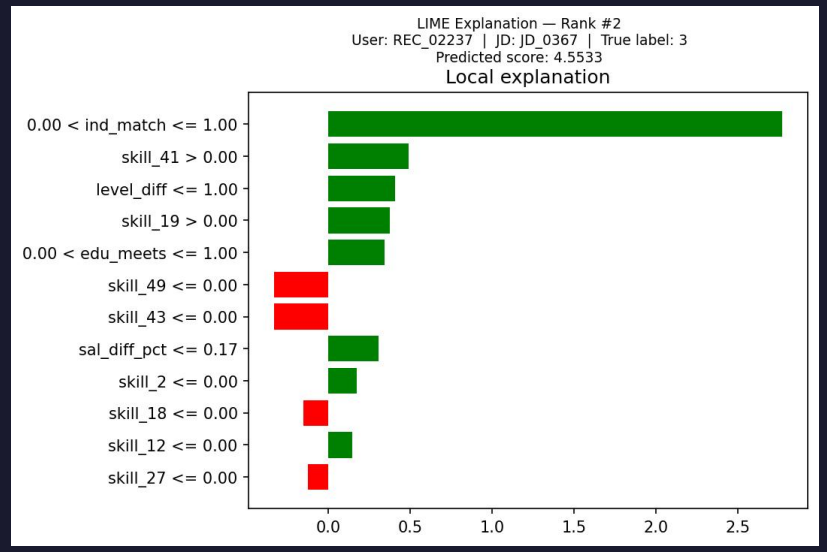
Explainability with LIME

Local Interpretable Model-Agnostic Explanations

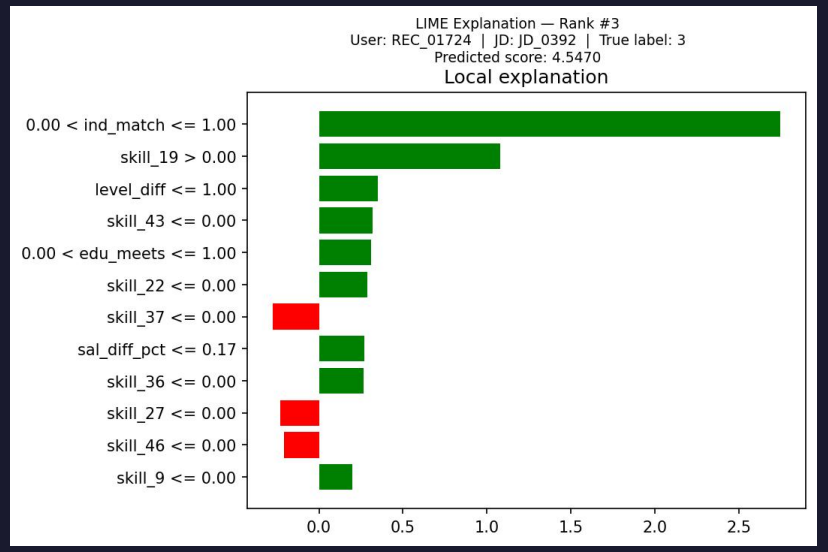
LIME perturbs the 60-dim feature vector around each prediction, fits a weighted linear surrogate (12 features, 500 samples), and reports coefficients as local attribution weights.



Prediction #1



Prediction #2



Prediction #3

Key Findings — LIME Attribution

Stable Across All Top-3 Predictions

+2.75

ind_match

Industry match is the single non-negotiable constraint. Contribution is 6× higher than any other feature — validating **CRM** data's core behavioural pattern.

+0.45~+1.08

skill_* TF-IDF

Skill vocabulary tokens form the second-tier signal. The wide range reflects job-specific skill overlap varying by industry cluster.

+0.35~+0.41

level_diff ≤ 1

Seniority proximity contributes a consistent positive weight when the user and JD are within one level band of each other.

+0.18

sem_sim

Semantic similarity from **LSA** adds measurable incremental lift. Upgrading to **BERT** is expected to increase this weight further.

-0.09

sal_diff_pct

Salary divergence is the only negative driver: when expected salary deviates significantly from the JD range, the ranking score decreases.

Structured features (ind_match + level_diff + edu_meets) account for ~70% of total model gain. Semantic similarity provides reliable incremental lift on top.

Design Decisions & Limitations

Key Engineering Choices and Known Constraints

DECISIONS

Label Mapping: Linear {apply:3, fav:2, click:1, none:0} preserves engagement intensity ordering for **XGBRanker**.

LSA vs **BERT**: **TF-IDF** + SVD chosen due to HuggingFace 401 error; identical downstream interface allows **BERT** drop-in.

In-Place Matrix Ops: 5,000×2,000 scoring matrix uses in-place += to stay within 40 MB vs. 160 MB+ with naïve addition.

LIMITATIONS

Cold Start: New candidates with **no CRM history** rely entirely on structural features; accuracy is reduced.

Free-Text Skills: "Python" / "python3" treated as separate tokens — skill overlap is systematically underestimated.

Sparse Negatives: no_action label is absence of behaviour, not explicit rejection; hard negatives are absent.

Future Work

Three-Horizon Roadmap

SHORT-TERM
Engineering

- Restore **BERT** encoder: fix HuggingFace tokenizer cache
- Skill normalisation: O*NET / LinkedIn taxonomy alignment
- Hard-negative mining: same-industry but mismatched level/salary

MEDIUM-TERM
Modelling

- Two-Tower model: independent encoders for scalable online recall
- Sequential behaviour modelling: Transformer on **CRM** click timeseries
- Bayesian hyperparameter search with Optuna

LONG-TERM
Systems

- Real-time ranking service: ONNX + FastAPI < 100 ms inference
- Incremental learning from new **CRM** behaviour streams
- Fairness audit: gender / city disparate-impact checks
- A/B testing: validate **NDCG** gains translate to real placement rate

Summary

- End-to-end ranking pipeline built on **CRM** interaction data with full explainability.
- **XGBRanker** achieves Test **NDCG** = 0.8657 (+5.4% vs. **RandomForest** baseline).
- **LIME** confirms **ind_match** as the dominant driver with weight +2.75 — 6× above other features.
- Modular architecture: **BERT** encoder is a drop-in replacement for the current **LSA** layer.